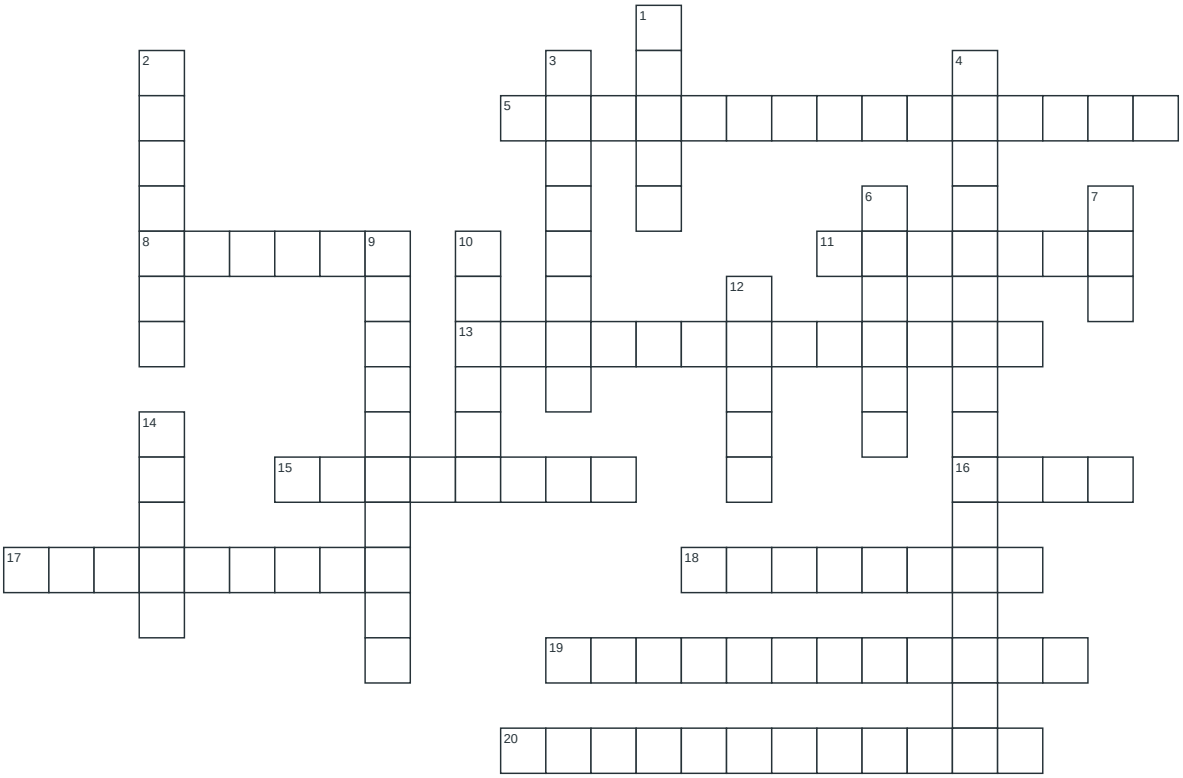
# Simple Machines Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



## ACROSS

**5.** SI-derived unit commonly used to measure speed or velocity.
**8.** Turning effect of a force about an axis.
**11.** Definition of the pivot point of a lever called
**13.** Simple machine is a sloped surface used to raise a load
**15.** Force that opposes relative motion between surfaces in contact.
**16.** State in which position does not change relative to a chosen reference point.
**17.** Orientation of motion, needed to describe a velocity vector.
**18.** The total length of a traveled path
**19.** Change in velocity per unit time.
**20.** The directed change from initial to final position

## DOWN

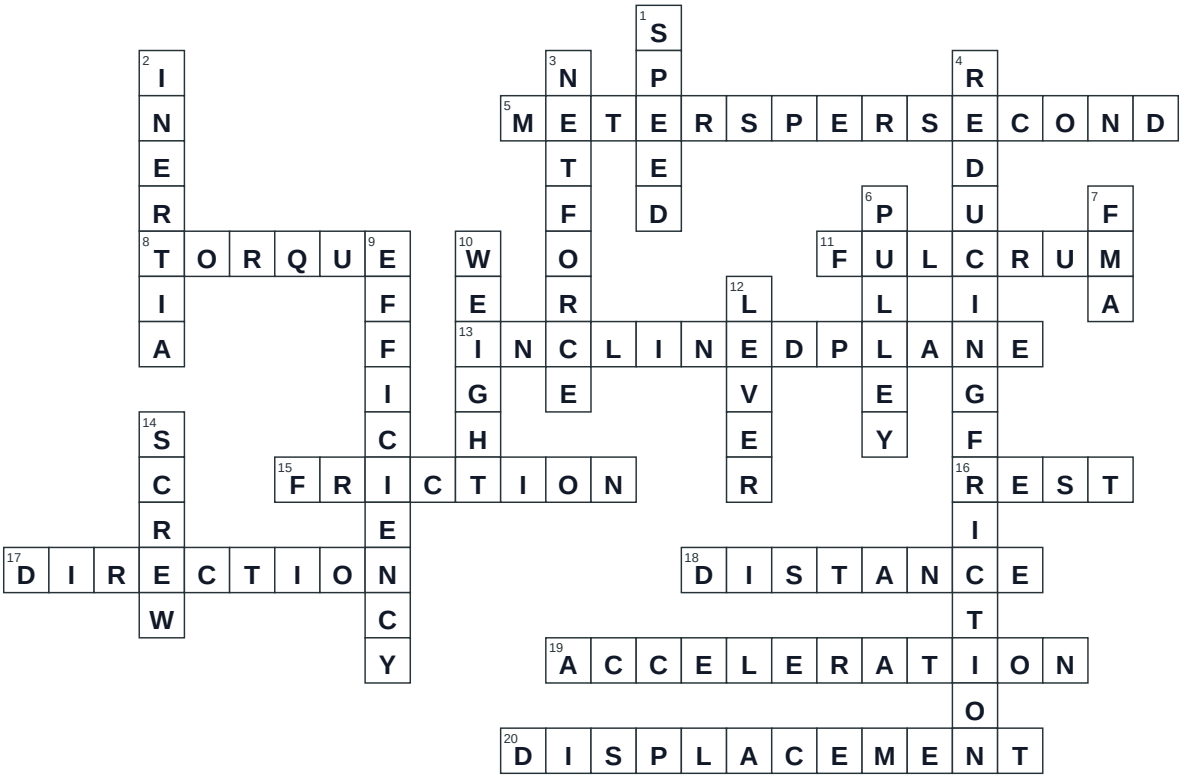
**1.** Quantity that is a scalar
**2.** Resistance to changes in motion
**3.** The vector sum of all forces acting on an object
**4.** Generally increases the fraction of input work delivered usefully
**6.** Simple machine consists of a grooved wheel guiding a rope
**7.** Relationship that expresses Newton's second law for constant mass
**9.** Useful output energy with total input energy
**10.** The gravitational force acting on an object
**12.** Simple machine uses a rigid bar that turns around a pivot
**14.** Simple machine is an inclined plane wrapped around a cylinder

# Simple Machines Crossword

Hard | 20 clues | Teacher Answer Key

### TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.




## ACROSS

**5.** Meters per second
**8.** Torque
**11.** Fulcrum
**13.** Inclined Plane
**15.** Friction
**16.** Rest
**17.** Direction
**18.** Distance
**19.** Acceleration
**20.** Displacement

## DOWN

**1.** Speed
**2.** Inertia
**3.** Net Force
**4.** Reducing Friction
**6.** Pulley
**7.** F = ma
**9.** Efficiency
**10.** Weight
**12.** Lever
**14.** Screw